

User Guide for the Canadian Income Survey Public Use Microdata File, 2022

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1. Introduction

Income data has been used extensively by researchers to better understand the economic well-being of Canadians. To meet the needs of these users, Statistics Canada has produced numerous cross-sectional public use microdata files (PUMFs). PUMFs for the Survey of Consumer Finance (SCF) were released until reference year 1997. With the end of the SCF, PUMFs for the Survey of Labour and Income Dynamics (SLID) were produced for reference years 1996 to 2011.

The Canadian Income Survey (CIS) was introduced for the 2012 reference year. The CIS is a cross-sectional survey developed to provide information on the income and income sources of Canadians, with their individual and household characteristics. It is a short questionnaire which is asked of a sub-sample of respondents to the Labour Force Survey (LFS), gathering information on labour market activity, school attendance, support payments, child care expenses, personal income, food security and characteristics and costs of housing. The CIS content is supplemented with information from the LFS on individual and household characteristics (e.g. age, educational attainment, main job characteristics, and family type) and with tax data for income and income sources (Statistics Canada, 2023a).

The CIS PUMF is an anonymized microdata file that contains only a subset of variables that are available on the CIS master file. Various techniques have been employed to protect CIS respondents against the risk of disclosure.

2. The CIS public use microdata file (PUMF)

2.1. Improvements to the Canadian Income Survey methodology

In recent years, the CIS has introduced improvements to the methods and sources used to produce income estimates. Beginning with the 2021 reference year, CIS income data were produced from the Administrative Personal Income Masterfile (APIM), a comprehensive source of personal income data generated not only from T1 tax returns, but also from associated tax slips. CIS estimates for previous years were produced using T1 tax returns only. Please consult the document [“Improvements to the Canadian Income Survey Methodology for the 2021 Reference Year”](#) for more information.

More improvements were introduced with the 2022 reference year. Among these improvements is an increase in the sample size for a large subset of the CIS content (this larger sample is referred to as “CIS-Plus”). However, the CIS PUMF only includes the CIS sample, not the CIS-Plus additional sample. The weighting methodology was also improved and the target population for income data was changed from persons aged 16 years and over to persons aged 15 years and over. Please consult the document [“Improvements to the Canadian Income Survey Methodology for the 2022 Reference Year”](#) for more information.

2.2. New variable

Starting in 2022, the CIS collects repayments of pandemic-related benefits that were received in error or for which the eligibility criteria were not met. These repayments are identified through T1 and T4 filings, which are among the sources used for the CIS estimates. The process of recording overpayments or

payments made to ineligible recipients of pandemic-related benefits in the CIS may take several years to complete.

In the CIS, for people who received Employment Insurance (EI) benefits in 2022, the repayment amount was deducted from EI benefits, while for people who did not receive EI benefits, the repayment was reported separately. In the CIS PUMF, a new variable called COVRET includes the repayment amount that was deducted from EI benefits, as well as the repayment amount that was reported separately. It should be noted that because the repayments represent a non-discretionary expense, they were deducted from disposable income when calculating the Market Basket Measure poverty estimates.

2.3. Occupation classification

The occupation classification has been updated. The CIS now uses National Occupational Classification (NOC) 2021 V1.0. Occupation variables were revised back to 2012.

2.4. File structure

The CIS PUMF is a “person file”, meaning that there is a record for each person in responding households. If the unit of analysis is not the person (for example the household, economic family or census family), users should retain only one record per unit of analysis. An easy way to do this is to keep only the record of the major income earner of the unit of analysis:

- if the unit of analysis is the economic family, keep only records where EFMJIE=1;
- if the unit of analysis is the household, keep only records where HHMJIE=1;
- if the unit of analysis is the census family, keep only records where CFMJIE=1.

Note that income data at the person level is only available for those aged 15 years or older.

Table 1: Number of records per unit of analysis in the 2022 PUMF

Unit of analysis	Number of records
All persons	96,302
Persons aged 15 years and over	81,163
Census families	45,622
Economic families	43,428
Households	41,978

2.5. Confidentiality of the public use microdata

The production of a public use microdata file includes many safeguards to prevent the identification of any one person. CIS uses a number of techniques to ensure confidentiality:

- Data reduction involves limiting the amount of identifying information on the PUMF. Techniques include removing direct identifiers, reducing the level of detail, grouping and suppressing data values for specific records.

- Data perturbation involves applying protective measures by coarsening or perturbing the data to hamper re-identification attempts. The addition of noise and data swapping are examples of perturbation techniques that are often employed.
- Quantitative variables with very large positive or negative values are usually rare or unique in the population. Such extreme values are often top-coded which involves replacing the top values while preserving the integrity of the file for the purposes of producing precise and accurate statistics.
- All income values are rounded.

2.6. Reserved codes

If the coverage of a variable does not extend to a certain population sub-group, then there are no valid values for that sub-group and the values (reserved codes) that do appear are in the form of 6, 96, 996, 999.6 and so on, which indicates that the variable is not applicable (valid skip). The coverage of each variable on the file is referred to in the data dictionary as the “universe”.

For certain records, no valid value is available, although the value is applicable. Such missing values appear with a reserved code such as 9 or 99.

It is important to account for reserved codes in any analysis, particularly with numeric variables. If the calculation of means or aggregates seems too high, users should check to ensure that reserved codes were excluded from the calculation.

3. Survey methodology

3.1. Target population

The survey is conducted nationwide, in both the provinces and the territories. It covers all individuals in Canada, excluding persons living on reserves and other Aboriginal settlements in the provinces, the institutionalized population, and households in extremely remote areas with very low population density. Overall, these exclusions amount to less than 2 percent of the population. However, users should note that although the territories are covered by the survey, they are not included in the PUMF.

3.2. Sample design

The CIS is a sample survey with a cross-sectional design which is administered to a sub-sample of LFS respondents. The LFS sample is drawn from an area frame and is based on a stratified, multi-stage design that uses probability sampling.

The LFS uses a rotating panel sample design. In the provinces, selected dwellings remain in the LFS sample for six consecutive months. Each month about one-sixth of the LFS sampled dwellings are in their first month of the survey, one-sixth are in their second month of the survey, and so on. These six independent samples are called rotation groups.

For the 2022 CIS, six rotation groups from the LFS were used for the provinces: the rotation groups answering the LFS for the last time in January to June of 2023. The CIS total sample size was about 60,000 households in the provinces.

3.3. Data collection

Responding to this survey is voluntary.

Data are collected directly from survey respondents and extracted from administrative files.

The LFS interview is completed by a responsible member of the household who generally provides the LFS responses for all members. Following the LFS interview, and subject to operational constraints, the interviewer then requests that this same member answers the CIS questionnaire. This person generally responds to the CIS questions for all household members aged 15 years or older (household members aged 16 years or older for reference years prior to 2022).

For the 2022 CIS, interviews were conducted by telephone or in person by Statistics Canada interviewers. In addition, respondents who met certain criteria were offered the option of completing the survey on-line.

The questionnaire follows standard practices and wording used in a computer-assisted interviewing environment, such as the automatic control of flows that depend upon answers to earlier questions and the use of edits to check for logical inconsistencies and capture errors.

In order to reduce response burden and improve the accuracy of the data, CIS does not ask respondents questions on every aspect of their income. Rather, CIS retrieves this information from tax records. CIS respondents are informed of these plans during the interview, a practice which is called informed replacement.

3.4. Data processing

The Canadian Income Survey is processed using the Social Survey Processing Environment (SSPE) processing system. The SSPE consists of a set of generalized processes to be used in the processing activities of the Survey Life Cycle. The purpose of these processes is to allow subject matter and survey support staff to specify and run the processing of a survey in a timely fashion with high quality outputs.

3.5. Imputation

Households are kept as respondents if information for at least one person in the household was provided, and any key data that is missing for individuals within responding households is imputed. Imputation is carried out for income variables as well as variables related to labour, school attendance, food security and housing.

CIS uses a nearest neighbour approach for the imputation of most income variables, and for labour, school attendance, food security and housing variables. This imputation method involves the selection of a donor record based on matching variables. First, a set of matching variables, each of which is correlated with the variables to be imputed, is defined. Then, through the combined use of a score function (for categorical matching variables) and a distance function (for numeric matching variables), the most similar consistent donor record is identified and used to impute data for the record.

Deterministic imputation is also used for selected income variables. Amounts for certain government programs, such as refundable provincial tax credits, are derived based on qualifying characteristics.

3.6. Estimation

The CIS sample is a sub-sample of the Labour Force Survey sample. LFS uses a complex random sampling plan to select the households. Each household in the sample represents a number of other households in the population. Estimates for a given characteristic are obtained by multiplying the survey weight by the corresponding value of this characteristic. The key step in the point estimation process is therefore the derivation of the weight.

The initial weights are the LFS sub-weights, which are then adjusted to account for the fact that the CIS is a sub-sample of the LFS sample.

Two types of adjustment are then applied to these weights in order to improve the reliability of the estimates. The weights are first inflated to compensate for CIS non-response. The non-response adjusted weights are then adjusted to ensure that estimates on relevant population characteristics respect population totals from sources other than the survey.

The first set of population totals used by the CIS are estimates provided by Statistics Canada's Demography Division of population counts based on the 2016 Census of Population. For each province, population counts by age/sex groups and economic family size are used. CIS also employs population counts for six Census Metropolitan Areas (Montreal, Toronto, Winnipeg, Calgary, Edmonton, and Vancouver).

The second set of totals is derived from the Administrative Personal Income Masterfile (APIM) and is intended to ensure that the weighted distribution of income (based on wages and salaries) in the data set matches that of the Canadian population.

For CIS 2020 and CIS 2021, two sets of totals were added to account for the changes in response patterns and new sources of income available due to the pandemic. The first set of totals, based on the 2021 Census results, adjusted the weights according to the proportion of households by housing tenure. The second set of totals, based on APIM estimates, adjusted the weights to reflect the number of Canada Emergency Response Benefit recipients and Canada Emergency Student Benefit recipients in 2020, and the number of Canada Recovery Benefit recipients in 2021.

For CIS 2022, the weight adjustment for the proportion of households by housing tenure remains. In addition, the weights are adjusted to reflect the number of families by economic family type and size, using estimates from Statistics Canada's Centre for Demography.

Further, the weights are adjusted to reflect the number of social assistance recipients. Control totals were created for the number of social assistance recipients by province from the APIM.

The CIS survey weights are then modified for the PUMF so that valid inferences to the CIS population can be made.

It is the responsibility of data users to apply the weights for any estimates they wish to produce. If weights are not used, the results derived from the microdata cannot be considered to be representative of the survey population. On the CIS PUMF file, the weight variable is named FWEIGHT.

3.7. Comparability over time

The CIS reports on many of the same statistics as the Survey of Labour and Income Dynamics (SLID), which last reported on income for the 2011 reference year. Prior to SLID, income data came from the Survey of Consumer Finances (SCF)¹.

The CIS uses a different methodology compared to that used in SLID. Data from the SLID were revised for the years 2006 to 2011 to allow CIS data to be compared with data for earlier years (Statistics Canada, 2015). However, SLID public use microdata files were not revised. As a result, it is not recommended to compare CIS results with those produced using the SLID PUMFs (Statistics Canada, 2014).

The improvements implemented with the 2021 and 2022 reference years also introduced breaks in some series. However, these changes had only a small impact on key CIS estimates and trends, while making significant improvements in data quality (Statistics Canada, 2023b; Statistics Canada, 2024).

4. Data quality

Survey errors come from a variety of different sources. They can be classified into two main categories: sampling errors and non-sampling errors.

4.1. Sampling errors

Sampling errors occur because inferences about the survey population are based on data from a sample of that population rather than the entire population. The sample design, the variability of the characteristic being measured and the sample size all contribute to the magnitude of the sampling error.

The standard error is a common measure of sampling error. It measures the degree of variation introduced in estimates by selecting one particular sample rather than another of the same size and design. Another widely used measure of sampling error is the coefficient of variation (CV) which is the estimated standard error expressed as a percentage of the estimate. Table 2 gives provincial- and national-level CVs for a number of CIS estimates for 2022. These CVs were computed using the bootstrap approach for estimates produced with the CIS master file. They are provided as an indicator of the sampling variability associated with selected estimates. Users should expect some differences due to the confidentiality measures applied to the PUMF data.

¹ There are a number of additional household surveys and statistical programs that collect and provide income data at Statistics Canada. They include the T1 Family File, the Census of Population, the 2011 National Household Survey, as well as surveys such as the Survey of Financial Security and the Longitudinal and International Study of Adults. These survey programs may collect income data directly from respondents through survey collection, or from income tax returns. Across the various data sources, there are differences with the methodologies, the concepts, the questions and the sources of the income data itself.

Table 2: Coefficients of variation for selected CIS variables^{1,2}, Canada and provinces, 2022

Province	Variables						
	Median total income	Median market income	Median wages and salaries	Median employment insurance benefits	Median social assistance	Median other income	Number of persons in poverty (MBM ³)
Newfoundland and Labrador	2.4	2.5	3.6	9.0	8.9	13.1	6.8
Prince Edward Island	3.0	3.2	3.9	9.3	25.0	17.7	10.5
Nova Scotia	1.6	2.8	2.7	5.3	12.0	8.5	7.4
New Brunswick	1.2	1.8	1.8	6.5	9.7	10.2	6.3
Quebec	1.5	1.7	1.3	4.7	12.7	7.3	4.7
Ontario	0.9	1.1	1.2	4.9	4.0	4.4	3.1
Manitoba	1.4	1.3	1.5	8.4	3.8	13.7	5.5
Saskatchewan	1.6	1.9	2.0	9.3	10.7	12.0	7.2
Alberta	1.5	2.1	2.5	12.3	15.8	13.5	6.9
British Columbia	1.4	1.5	1.8	4.4	15.3	8.2	4.4
Canada	0.5	0.7	0.8	3.1	1.7	1.3	1.9

Notes:

1. Coefficients of variation were computed for estimates produced with the CIS master file.
2. All estimates are at the economic family-level except the number of persons in poverty.
3. Market Basket Measure, 2018 base

4.2. Non-sampling errors

Non-sampling errors resulting from human errors such as simple mistakes, misunderstanding or misinterpretation will generally have a minor impact on the overall accuracy of the estimates. Errors occurring systematically and errors arising from sources such as coverage, erroneous response, non-response and processing can have, on the other hand, a major impact on the reliability of estimates. Considerable time and effort is invested into reducing non-sampling errors in CIS.

4.2.1. Coverage error

Coverage error arises when sampling frame units do not exactly represent the target population. Units may have been omitted from the sampling frame (undercoverage), or units not in the target population may have been included (overcoverage), or units may have been included more than once (duplicates). Undercoverage represents the most common coverage problem. Slippage is a measure of survey

coverage error. It is defined as the percentage difference between control totals (postcensal population estimates) and weighted sample counts. In 2022, the CIS person-level slippage rate was 10.2%.

4.2.2. Response error

Response or measurement errors occur when the response provided differs from the real value; such errors may be attributable to the respondent, the interviewer, the questionnaire, the collection method or the respondent's record-keeping system. Such errors may be random or they may result in a systematic bias if they are not random.

Several measures are taken to reduce the level of response error. These measures include questionnaire review and testing using cognitive methods, focus groups, the use of highly skilled interviewers, extensive training of interviewers with respect to the survey procedures and content, and the observation and monitoring of interviewers to detect problems of questionnaire design or misunderstanding of instructions.

4.2.3. Non-response error

Non-response errors result from a failure to collect complete information on all units in the selected sample.

Non-response produces errors in the survey estimates in two ways. First, non-respondents often have different characteristics from respondents, which can result in biased survey estimates if non-response is not corrected properly. Secondly, it reduces the effective size of the sample, since fewer units than expected answered the survey. As a result, the sampling variance increases and the precision of the estimate decreases.

Total non-response is dealt with by adjusting the weights of the respondents to account for the non-respondents. In 2022, the CIS final response rate was 70.1%

4.2.4. Processing error

Processing error is the error associated with activities conducted once survey responses have been received. It includes all data handling activities after collection and prior to estimation. CIS is processed using the SSPE system and a series of verifications take place to ensure that no errors are introduced.

5. Guidelines for release

5.1 Minimum sizes of estimates for release

Suppression rules, or data reliability cut-offs, are established based on the sample size that underlies the estimate. In general, a sample size of 25 observations is required for the estimate to be published.

Depending on the type of estimate, this rule can vary slightly. These rules help protect the confidentiality of survey respondents and ensure the reliability of estimates.

Table 3: Suppression rules

Estimate	Suppress if:
Percentage, Distribution, Proportion/Share	
• % under the low income cut-off • Income distribution	Denominator sample size* < 25 or Denominator sample size* < 100 and numerator sample size < 5
Deciles (shares, means and upper income limits)	
• Shares of income by decile • Average income by decile • Upper income limits	Sample size/10 < 25 or Upper income limit for upper income decile or total of deciles
Other estimates	
• Counts • Averages • Medians • Aggregate income • Gini coefficients	Sample size < 25

*The denominator sample size refers to the sample size of the total estimate from which the distribution, percentage, proportion or share is derived.

5.2 Rounding guidelines

To ensure that estimates from this microdata file intended for publication or any other type of release correspond to estimates that would be obtained by Statistics Canada, users should comply with the following guidelines for rounding estimates:

- Estimates in the body of a statistical table must be rounded to the nearest hundredth using the traditional rounding technique. If the first or only number to be eliminated is between 0 and 4, the preceding number does not change. If the first or only number to be eliminated is between 5 and 9, the value of the last number to be retained increases by 1. For example, when using the traditional technique of rounding to the nearest hundredth, if the last two numbers are between 00 and 49, they are replaced by 00 and the preceding number stays as is. If the last two numbers are between 50 and 99, they are replaced by 00 and the preceding number increases by 1.
- Total partial sub-totals and total sub-totals in statistical tables must be calculated using their unrounded corresponding components, then rounded in turn to the closest hundredth using the traditional rounding technique.
- Means, ratios, rates and percentages must be calculated using unrounded components (i.e., numerators and/or denominators), and then rounded to a decimal using the traditional rounding technique.

- d) Totals and differences in aggregates (or ratios) must be calculated using unrounded components, then rounded to the nearest hundredth (or decimal place) using the traditional rounding technique.
- e) Unrounded estimates should not be published or released. Unrounded estimates give the impression that they are much more precise than they actually are.

6. Notes and definitions

6.1. General concepts

6.1.1. Reference year of the survey

Corresponds to the calendar year from January 1st to December 31st prior to the collection period.

6.1.2. Units of analysis

6.1.2.1. Household

A household is defined as any person or group of persons living in a dwelling. A household may consist of any combination of: one person living alone, one or more families, a group of people who are not related but who share the same dwelling.

6.1.2.2. Economic family

An economic family is defined as a group of two or more persons who live in the same dwelling and are related to each other by blood, marriage, common law, adoption or a foster relationship. A person not in an economic family is a person living either alone or with others to whom he or she is unrelated, such as roommates or a lodger.

"Economic family type" refers to either economic families or persons not in an economic family. It is set using the major income earner in the economic family as the reference person. Table 4 lists the economic family types and groupings.

Table 4: Economic family type (classification)

Economic families

- Elderly families
 - Elderly couples
 - Other elderly families
- Non-elderly families
 - Couples
 - Couples with children
 - Couples with other relatives
 - Lone-parent families
 - Male lone-parent families
 - Female lone-parent families
 - Other non-elderly families

Persons not in an economic family

- Elderly males
- Elderly females
- Non-elderly males
- Non-elderly females

6.1.2.3. Census family

"Census family type" refers to either census families or persons not in a census family. The term "census family" corresponds to what is commonly referred to as a "nuclear family" or "immediate family". In general, it consists of a married couple or common-law couple with or without children, or a lone-parent with a child or children. Furthermore, each child does not have his or her own spouse or child living in the dwelling. Grandchildren living with their grandparent(s) but with no parents present also constitute a census family.

Persons "not in a census family" are those living alone, living with unrelated individuals, or living with relatives but not in a spouse or parent-unmarried child relationship.

By definition, all persons who are members of a census family are also members of the same economic family. Table 5 lists the census family types and groupings.

Table 5: Census family type (classification)

Census families

- Couples
- Couples with children aged 24 and under
- Couples with children aged 25 and over only
- Lone-parent families
 - Male lone-parent families with children aged 24 and under
 - Male lone-parent families with children aged 25 and over only
 - Female lone-parent families with children aged 24 and under
 - Female lone-parent families with children aged 25 and over only

Persons not in a census family

- In a one-person economic family
- In a multi-person economic family

6.1.3. Household and family concepts

6.1.3.1. Major income earner

This characteristic is important for the derivation of detailed family and household types. For each household and family, the major income earner is the person with the highest income before tax.

For persons with negative total income before tax, the absolute value of their income is used, to reflect the fact that negative incomes generally arise from losses "earned" in the marketplace which are not meant to be sustained. In the rare situations where two persons have exactly the same income, the older person is the major income earner.

6.1.3.2. Elderly families

The major income earner is aged 65 or over.

6.1.3.3. Non-elderly families

The major income earner is under age 65.

6.1.3.4. Elderly persons

Persons 65 years or older.

6.1.3.5. Non-elderly persons

Persons less than 65 years old.

6.1.3.6. Couples/spouses

Couples, including legally married, common-law and same-sex relationships, where one of the spouses is the major income earner.

6.1.3.7. Children

A child or children (by birth, adopted, step, or foster) of the major income earner. For census families, foster children are not included. For economic families, children are defined as being under age 18. Other relatives may also be in the family.

6.1.3.8. Lone-parent families

Includes at least one child as defined above. Families where the parent is 65 or older are excluded.

6.1.3.9. Relative

A person related to the major income earner by blood, marriage, adoption or common-law.

6.1.3.10. Other relative in economic families

A person in the economic family who is not the major income earner nor his/her spouse or child under age 18.

6.2. Income

6.2.1. The concept of income

In the context of persons, households, census families and economic families, total income refers to monetary receipts from certain sources, before income taxes and deductions, during the reference year. It includes employment income from wages, salaries, tips, commissions and net income from self-employment (for both unincorporated farm and non-farm activities); income from government sources, such as social assistance, child benefits, employment insurance, Old Age Security pension, Canada or Quebec pension plan benefits and disability income; income from employer and personal pension sources, such as private pensions and payments from annuities and Registered Retirement Income Funds; income from investment sources, such as dividends and interest on bonds, accounts, Guaranteed Income Certificates and mutual funds; and other regular cash income, such as child support payments received, spousal support payments (alimony) received and scholarships. The monetary receipts included are those that tend to be of a regular and recurring nature. It excludes one-time receipts, such as lottery winnings, gambling winnings, cash inheritances, lump sum insurance settlements, capital gains and Registered Retirement Savings Plan (RRSP) withdrawals. Capital gains are excluded because they are not by their nature regular and recurring. It is further assumed that they are less likely to be fully spent in the period in which they are received, unlike income that is regular and recurring. Also excluded are employer's contributions to registered pension plans, Canada and Quebec pension plans, and employment insurance. Finally, voluntary inter-household transfers, imputed rent, goods and services produced for barter, and goods produced for own consumption are excluded from this total income definition.

6.2.2. Classification of Income

Market Income

- Earnings/Employment income
 - Wages, salaries and commissions
 - Self-employment income
 - Farm
 - Non-farm
- Investment income
- Retirement pensions
- Other income

(plus) Government transfers

- Child benefits
- Canada Pension Plan (CPP)/Quebec Pension Plan (QPP) benefits
- Old Age Security (OAS) and Guaranteed Income Supplement (GIS)/Spouse's Allowance
- Employment Insurance benefits
- Social assistance
- Workers' compensation benefits
- GST/HST credit
- Canada workers benefit
- Federal refundable tax credits

- Provincial tax credits and seniors' supplements
- COVID-19 benefits

(equals) Total income

(minus) Income taxes

(equals) After-tax income

(minus) Non-discretionary expenses

- Employment Insurance contributions
- CPP/QPP contributions
- Registered pension plan contributions
- Union dues and professional membership dues and malpractice liability insurance premiums
- Child care expenses incurred in order to hold a paid job
- Support payments paid
- Direct medical expenses, including private insurance premiums
- Federal COVID-19 benefits repayment

(plus) Tenure type adjustment

(plus) Capital gain tax adjustment

(equals) Disposable income (2018-base Market Basket Measure)

6.2.3. Components of income

6.2.3.1. Market income

Market income is the sum of earnings (from employment and self-employment), investment income, (private) retirement income, and the items under "Other income".

6.2.3.2. Earnings/Employment income

This includes earnings from both paid employment (wages and salaries) and self-employment.

6.2.3.3. Wages, salaries and commissions

These are gross earnings from all jobs held as an employee, before payroll deductions such as income taxes, employment insurance contributions or pension plan contributions. Wages and salaries include the earnings of owners of incorporated businesses, although some amounts may instead be reported as investment income. Commission income received by salespersons as well as occasional earnings (for example for baby-sitting, delivering papers and cleaning) are included. Overtime pay is included.

6.2.3.4. Self-employment income – farm and non-farm

This is net self-employment income after deduction of expenses. Negative amounts (losses) are accepted. It includes income received from self-employment, in partnership in an unincorporated business, or in independent professional practice. Income from roomers and boarders (excluding that

received from relatives) is included. Note that because of the various inclusions, receipt of self-employment income does not necessarily mean the person held a job.

Self-employment income is subdivided into farm self-employment income and non-farm self-employment income. Farm self-employment income is reported by individuals who operate their own or a rented farm, either on own account or in partnership. Included are money receipts from the sale of farm products as well as related supplementary and assistance payments from governments. Income in kind is excluded.

6.2.3.5. Investment income

This includes interest received on bonds, deposits and savings certificates from Canadian or foreign sources, dividends received from Canadian and foreign corporate stocks, cash dividends received from insurance policies, net rental income from real estate and farms, interest received on loans and mortgages, regular income from an estate or trust fund and other investment income. Realized capital gains from the sale of assets are excluded. Negative amounts are accepted.

6.2.3.6. Retirement income

This is income from private sources such as employer pension plans and RRSPs; it may also be referred to as pension income. Amounts withdrawn from RRSPs at ages 65 or older are considered retirement income. Withdrawals before the age of 65 are not considered retirement income.

6.2.3.7. Other income

This sub-total includes all items of market income not included elsewhere. Among them are support payments received (also called alimony and child support) and income from a Registered Disability Savings Plan (RDSP). Items included on line 130 of the T1 tax return are also components of “other income”. These include, but are not restricted to, retirement allowances (severance pay/termination benefits), scholarships, lump-sum payments from pensions and deferred profit-sharing plans received when leaving a plan, the taxable amount of death benefits other than those from CPP or QPP, and supplementary unemployment benefits not included in wages and salaries.

Beginning with reference year 2021, T4A slips (Statement of Pension, Retirement, Annuity, and Other Income) were used to complement information from T1 tax returns, which better capture a few sources of other income (e.g., scholarships, bursaries, and RESP payments).

6.2.3.8. Government transfers

Government transfers include all direct payments from federal, provincial and municipal governments to individuals or families. It should be noted that many features of the tax system also carry out social policy functions but are not government transfers per se. The tax system uses deductions and non-refundable tax credits, for example, to reduce the amount of tax payable, without providing a direct income.

6.2.3.9. Child benefits

To be eligible for child benefits, a person must have the primary responsibility for the care and upbringing of one or more children under the age of 18. Most benefits are calculated by setting a maximum amount per family or per child and reducing that total by a certain percentage of the family's net income.

The following non-taxable benefits are included with child benefits:

Federal Programs

- Canada Child Benefit

Provincial Programs

- Newfoundland and Labrador Child Benefit
- Newfoundland and Labrador Mother Baby Nutrition Supplement
- Nova Scotia Child Benefit
- New Brunswick Child Tax Benefit
- New Brunswick Working Income Supplement
- New Brunswick School Supplement
- Quebec Family Allowance
- Ontario Child Benefit
- Manitoba Child Benefit
- Alberta Child and Family Benefit
- British Columbia Family Bonus
- British Columbia Earned Income Benefit
- British Columbia Child Opportunity Benefit

Beginning with reference year 2021, child benefit amounts were retrieved directly from Canada Child Benefit administrative files, while prior to 2021, child benefits were derived using a deterministic method. However, if no amount was found on tax files for families with children in the CIS, an amount was derived using a deterministic method. Conversely, if an amount was found for a person not in an economic family or in a family without children, the amount was set to zero.

6.2.3.10. Canada Pension Plan (CPP) and Quebec Pension Plan (QPP)

The CPP and QPP are compulsory contributory social insurance programs that provide a source of retirement income and protect workers and their families against loss of income due to disability or death.

6.2.3.11. Old Age Security (OAS) and Guaranteed Income Supplement (GIS)

The OAS pension is targeted to Canadian residents aged 65 and over. OAS recipients with income below a set threshold may also receive the federal GIS; and their spouses or widow(er)s, if aged 60 to 64 (and not yet eligible for OAS and GIS themselves), receive the Allowance or the Allowance for the Survivor.

In July 2022, a 10% increase in OAS pensions was introduced for recipients aged 75 years or older. For those turning 75 after July 1st, 2022, the increase begins the month following the 75th birthday.

6.2.3.12. Employment Insurance

Employment Insurance (EI) is a federal program which includes the following types of benefits: regular unemployment benefits, sickness benefits, maternity and parental benefits, and benefits for persons taking approved training courses or participating in job creation or job-sharing projects. To qualify, the claimant must have ceased receiving employment income and have worked a minimum number of weeks or hours of insurable employment over the preceding period.

6.2.3.13. Social assistance

Social assistance covers many provincial and municipal income supplements to individuals and families. It is usually provided only after all other possible sources of support have been exhausted.

Before the 2021 reference year, to complement information from T1 tax returns, a social assistance amount was imputed for persons who had reported receiving social assistance in the CIS. However, this question was often misunderstood, which led to a higher estimate for the number of social assistance recipients. Beginning with the 2021 reference year, T5007 slips were used to complement T1 information. In addition, provincial seniors' supplements are no longer included in "social assistance".

6.2.3.14. Workers' compensation

Workers' compensation is provided to protect all full-time and part-time employees from loss of salary due to work accidents or occupational diseases and help them to pay their medical expenses and other costs.

6.2.3.15. Goods and Services Tax (GST)/Harmonized Sales Tax (HST) credit

This credit is intended to offset the GST/HST for lower income families and individuals. In provinces where the administration of the GST is combined with the Provincial Sales Tax (PST), it is called the Harmonized Sales Tax Credit.

In 2022, there was a one-time payment to double the GST credit amount eligible individuals and families receive for a six-month period. Those entitled to receive the GST credit in October 2022 received this payment.

6.2.3.16. Canada workers benefit

The Canada workers benefit is a refundable tax credit to help individuals and families who are working and earning a low income.

6.2.3.17. Federal refundable tax credits

For 2022, they include the Climate Action Incentive Payment, the Eligible Educator School Supply Tax Credit, the Canada Training Credit, the Canadian Journalism Labour Tax Credit (CJLTC), and the Return of Fuel Charge Proceeds to Farmers tax credit, and the Refundable Medical Expense Supplement.

The Climate Action Incentive Payment applies to residents in Ontario, Manitoba, Saskatchewan, and Alberta where the federal carbon pollution pricing system is in effect. Payments are determined based on family type (single adults, couples, children) and are distributed quarterly. Payments for 2022 were up to \$279.95 per single adult with reduced amounts for the second adult and children in the same family. An additional increase of 10% was available to residents in small and rural communities.

The Eligible Educator School Supply Tax Credit allows eligible educators to claim up to \$1,000 of eligible teaching supply expenses.

The Canada Training Credit can be used for tuition and eligible fees paid for courses taken. Eligible individuals may accumulate \$250 per year. In addition, members of a qualified journalistic organization partnership are eligible to apply for the CJLTC awarded by a partnership.

The Return of Fuel Charge Proceeds to Farmers tax credit allows eligible self-employed farmers or individuals who are members of a partnership operating a farming business with one of more permanent establishments in Ontario, Manitoba, Saskatchewan, or Alberta to have a portion of fuel charge proceeds returned to them.

The Refundable Medical Expense Supplement is available to working individuals whose income is above the minimum threshold of \$3,841, with high medical expenses and adjusted family incomes of no more than \$55,449. Eligible persons can receive up to \$1,316 towards reimbursement for medical expenses accrued throughout the year.

6.2.3.18. Provincial tax credits and seniors' supplements

Provincial tax credits include most refundable provincial tax credits paid through the tax system plus several that are paid out separately and selected provincial income supplements for seniors.

Provincial credits include:

- Newfoundland and Labrador Income Supplement
- Newfoundland and Labrador One-Time Cost of Living Relief
- Nova Scotia Affordable Living Tax Credit
- Nova Scotia Poverty Reduction Tax Credit
- Prince Edward Island Harmonized Sales Tax Credit
- New Brunswick Harmonized Sales Tax Credit
- Quebec Solidarity Tax Credit
 - o Quebec Sales Tax Credit Component
 - o Housing Component
- Quebec Cost of Living Tax Credit

- Quebec Senior Assistance Tax Credit
- Ontario Trillium Benefit
 - o Ontario Sales Tax Credit
 - o Ontario Energy and Property Tax Credit
 - o Northern Ontario Energy Credit
- Ontario Seniors Homeowner Tax Grants
- Ontario Catch-up Payments
- Manitoba Tax Credit for Families
- Manitoba Affordability Tax Credit for Seniors
- Saskatchewan Low-Income Tax Credit
- Saskatchewan Affordability Tax Credit
- Alberta Climate Leadership Adjustment Rebate
- British Columbia Climate Action Tax Credit

Provincial seniors' supplements include:

- Newfoundland and Labrador Seniors' Benefit
- New Brunswick Low-Income Seniors' Benefit
- Ontario Guaranteed Annual Income System
- Manitoba 55 Plus
- Saskatchewan Seniors' Income Plan
- Alberta Seniors' Benefit Plan
- British Columbia Seniors' Supplement

6.2.3.19. COVID-19 benefits

In 2022, the majority of the government transfers to support those affected by the COVID-19 pandemic had been phased out previously or were subsequently phased out in 2022.

Federal emergency and recovery benefit programs available for part of 2022 include:

- Canada Recovery Caregiving Benefit
- Canada Recovery Sickness Benefit
- Canada Worker Lockdown Benefit

COVID-19 benefits also include various programs administered by provincial governments.

Total income

Total income refers to income from all sources including government transfers before deduction of federal and provincial income taxes. It may also be called income before tax (but after transfers). All sources of income are identified as belonging to either market income or government transfers.

6.2.3.20. Income tax

Income tax is the sum of federal and provincial income taxes payable (accrued) for the taxation year. Income taxes include taxes on income, capital gains and RRSP withdrawals, after taking into account exemptions, deductions, non-refundable tax credits, and the refundable Quebec abatement. The data are either taken directly from administrative records or estimated based on administrative records.

6.2.3.21. After-tax income

After-tax income is total income, which includes government transfers, less income tax. It may also be called income after tax.

6.2.3.22. Disposable income

Disposable income is used in the Market Basket Measure to determine the poverty status. It is income after deducting not only income taxes but also several non-discretionary expenditures. These expenditures are Employment Insurance, Canada Pension Plan, Quebec Pension Plan and Registered Pension Plan contributions, union dues (including professional membership dues and malpractice liability insurance premiums), child care expenses incurred in order to hold a paid job, support payments paid and direct medical expenses including private insurance premiums. Starting in 2022, the CIS collects repayments of pandemic-related benefits that were received in error or for which the eligibility criteria were not met. Because the repayments represent a non-discretionary expense, repayments made in 2022 were deducted from disposable income.

To recognize the lower shelter costs faced by homeowners without a mortgage, the 2008-base MBM made an adjustment to disposable income, which was called the Mortgage-Free Advantage (MFA). The MFA added the difference between the cost of the shelter component of the MBM basket (based on renters) and the typical shelter costs for mortgage-free homeowners, to the disposable income of mortgage free homeowners, placing homeowners without a mortgage on a more equal footing with renters in regard to their shelter costs, when evaluating their poverty status. For the 2018-base MBM, the MFA concept was extended to homeowners with a mortgage and subsidized renters; thereby, putting all families' shelter costs on a more equal footing when determining poverty estimates, regardless of their tenure type. This concept is referred to as the Tenure Type Adjustment (TTA).

The capital gain tax adjustment was also introduced in the 2018-base MBM. The 2008-base MBM did not include the values of capital gains or losses in disposable income, but it did deduct out of disposable income the taxes paid on capital gains. To address this inconsistency, capital gains taxes are no longer deducted from pre-tax income in the 2018-base MBM. This prevents families with large capital gains (and therefore large capital gains taxes) from appearing to be in poverty.

6.2.4. Family and household income

Family income (either economic family or census family) is the sum of the incomes of each person aged 15 years or older in the family as defined above. Household income is likewise the sum of the incomes of all persons aged 15 years or older in the household. Family and household membership is defined at a

particular point in time, while income is based on the entire calendar year. The family members or "composition" may have changed during the reference year, but no adjustment is made to family income to reflect this change.

6.3. Low income

Statistics Canada currently measures low-income using three low income lines: the Low Income Cut-offs (LICOs), the Low Income Measures (LIMs), and the Market Basket Measure (MBM). A research paper provides a description of the methods used to arrive at each of these thresholds. It also explains how low-income status and various low-income statistics are determined (Statistics Canada, 2016).

6.3.1. Low income measures (LIMs)

In simple terms, the LIM is a fixed percentage (50%) of median adjusted household income, where "adjusted" indicates that household needs are taken into account. Adjustment for household sizes reflects the fact that a household's needs increase as the number of members increases. Most would agree that a household of six has greater needs than a household of two, although these needs are not necessarily three times as costly. See *Section 6.4.6 – Adjusted incomes for family or household size* for more detail.

6.3.2. Low income cut-offs (LICOs)

The LICOs are income thresholds below which a family will likely devote a larger share of its income on the necessities of food, shelter and clothing than the average family. The approach is essentially to estimate an income threshold at which families are expected to spend 20 percentage points more than the average family on food, shelter and clothing². According to the most recent base for LICOs, the 1992 Family Expenditures Survey, the average family spent 43% of its after-tax income on food, shelter and clothing.

6.3.3. Market basket measure (MBM)

In August 2018, the MBM was set as Canada's first official poverty line. The MBM is based on the cost of a specific basket of goods and services representing a modest, basic standard of living. It includes the costs of food, clothing, footwear, transportation, shelter and other expenses for a reference family of two adults aged 25-49 and two children (aged 9 and 13). It provides thresholds for a finer geographic level than the LICO, allowing, for example, different costs for rural areas in the different provinces. These thresholds are compared to disposable income of families to determine the poverty status.

Between 2018 and 2020, Statistics Canada and Employment and Social Development Canada (ESDC) conducted a comprehensive review of the MBM to ensure that the basket continues to reflect a modest, basic standard of living, that the costs of the contents of the basket in different geographical regions are estimated as precisely as possible, and that the income available to families to purchase the basket is appropriately defined and measured (Statistics Canada, 2020).

² Twenty percentage points are used based on the rationale that a family spending 20 percentage points more than the average would be in "straitened circumstances".

The re-based MBM – referred to as the “2018-base MBM” – updated the 2008-base MBM developed between 2008 and 2010. Users should be aware that this microdata file uses the 2018-base MBM methodology.

6.3.4. Low income rate and low income gap ratio

To determine whether a person is in low income, the appropriate low income line (LIL) is compared to the income of the person’s economic family (or household)³. If their income is below the LIL, the individual is considered to be in low income. In other words, “persons in low income” should be interpreted as persons who are part of low income families (or households), including persons living alone whose income is below the LIL. Similarly, “children in low income” means “children who are living in low income families (or households)”. Overall, the low income rate for persons can then be calculated as the number of persons in low income divided by the total population. The same can be done for various sub-groups of the population; for example, low income rates by age, sex, or province.

After having determined that an individual is in low income, the low income gap ratio can be calculated by using the amount that the person’s family (or household) income falls short of the LIL, expressed as a percentage of the LIL. For example, an individual living in a family (or household) with an income of \$15,000 and a LIL of \$20,000 would have a low income gap of \$5,000. In percentage terms, the “gap ratio” would be 25%⁴. The average gap ratio for a given population is the average of these values as calculated for each person.

6.4. Analytical concepts

6.4.1. Zero income

Users should keep in mind whether to include or exclude persons or families who did not receive a given type of income in their calculations. In the case of total family income, the difference from including or excluding units with zero income is small since there are very few families that do not receive any income. However, if one is interested in the average amount of individual self-employment earnings, the value will be quite different if persons who did not receive that type of income are included or not.

6.4.2. Negative values

Negative income amounts can arise in two ways: net losses from self-employment (expenses exceed receipts), or net investment losses (losses exceed gains). As with zero values, negative values can have a large impact on results. In general, negative values are treated no differently than positive values, but there are a few exceptions: for the calculation of both Gini coefficients and the low income gap, negative values are converted to zeroes; and in the derivation of the major income earner of a family or household, the absolute value is used instead.

6.4.3. Median income

The median income is the value for which half of the units in the population have lower incomes and half has higher incomes. To derive the median value of income, units are ranked from lowest to highest

³ When using the LICO or the MBM, the economic family is the appropriate unit. When using the LIM, the household is the appropriate unit.

⁴ For the calculation of this low income gap, negative incomes are treated as zero.

according to their income and then separated into two equal-sized groups. The value that separates these groups is the median income (50th percentile).

Because the median corresponds exactly to the midpoint of the income distribution, it is not, contrary to the mean, affected by extreme income values. This is a useful feature of the median, as it allows one to abstract from unusually high values held by relatively few people.

Since income distributions are typically skewed to the left - that is, concentrated at the low end of the income scale - median income is usually lower than mean income.

6.4.4. Average or mean income

The average or mean income is computed as the total or "aggregate" income divided by the number of units in the population. It offers a convenient way of tracking aggregate income while adjusting for changes in the size of the population.

There are two drawbacks to using average income for analysis. First, since everyone's income is counted, the mean is sensitive to extreme values: unusually high income values will have a large impact on the estimate of the mean income, while unusually low ones, i.e. highly negative values, will drive it down. Secondly, it does not give any insight into the allocation of income across members of the population. To examine allocation of income, measures such as percentiles or Gini coefficients may be used.

6.4.5. Percentiles

Income percentiles, like quintiles and deciles, are a convenient way of categorizing units of a given population from lowest income to highest income for the purposes of drawing conclusions about the relative situation of people at either end or in the middle of the scale. Rather than using fixed income ranges, as in a typical distribution of income, it is the fraction of each population group that is fixed.

First, all the units of the population, whether individuals or families, are ranked from lowest to highest by the value of their income of a specified type, such as after-tax income. Then the ranked population is divided into five groups of equal numbers of units, called quintiles. Analogously, dividing the population ranked by income into ten groups, each comprising the same number of units, produces deciles.

Most analyses should be carried out on the people of different percentiles within one population distribution. Care should be taken in making comparisons between percentiles that resulted from different distributions, because any difference in either the population or the income concept used to rank units could have a large effect. It is probable that both the income ranges represented by each percentile and the people making up each percentile will be different.

6.4.6. Adjusted incomes for family or household size

When comparing grouping unit (family or household) incomes to study such things as income adequacy or socio-economic status, one often wants to take the unit size and composition into account. The income amount itself is not sufficient to understand a unit's financial well-being without knowing how many people are sharing it. In general, two approaches have been used to help with the analysis of grouping unit income. One is to produce data by detailed unit types, so that within a given type,

differences in unit size are less significant. The other way to take into account unit size and composition is to adjust the income amount by an adjustment factor.

The simplest method is to use per capita income, that is, to divide the grouping unit income by the number of members it includes. A limitation of per capita income, however, is that it tends to underestimate economic well-being for larger units as compared to smaller units. This is due to the fact that it assumes equal living costs for each member of the unit, but some costs, primarily those related to shelter, decrease proportionately with unit size (they may also be lower for children than for adults). For example, the shelter costs for an adult married couple with no children are arguably not much more than those for an adult living alone.

To take such economies of scale into account, it is common to use an "equivalence scale" to adjust grouping unit incomes. Adjusted income is obtained by dividing family or household income by the square root of the number of members in the family or household.

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